

WASP-43b

R-1379

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_211227 · created 2026-08-01T01:43:01+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459575.9058 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.193 +/- 0.011
Transit depth (Rp/Rs)^2	3.71 +/- 0.44 [%]
Orbital Inclination (inc)	86.24 +/- 1.34
Airmass coefficient 1 (a1)	0.954 +/- 0.019
Airmass coefficient 2 (a2)	0.026 +/- 0.011
Scatter in the residuals of the lightcurve fit is	2.03 %
Best Comparison Star	#3 - [472, 270]
Optimal Method	PSF photometry
Transit Duration (day)	0.0612 +/- 0.0019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.193 ± 0.011 vs 0.1594 (deviation 3.05σ)
Duration fit vs published	0.0612 d vs 0.0512 d (deviation 5.24σ)
Mid-time O–C	5.2 min
Depth SNR	17.5
Residual scatter	2.03 %

Light curve

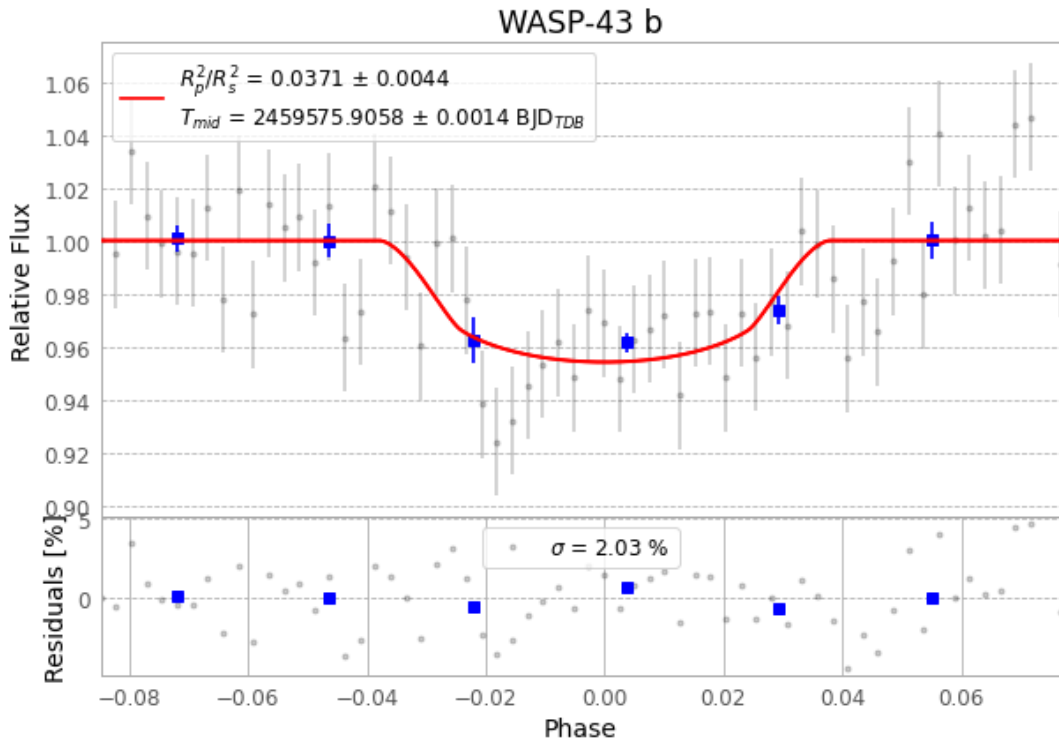


Figure 1 — FinalLightCurve_WASP-43 b_2021-12-27.png (SHA-256 f026a5bec7c1b56c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [452, 143], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ea61d00e60b49fea...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

64 FITS frames (42 MB), WASP-43211227080012.FITS → WASP-43211227110912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-211227.log (df121a3e963c...), FinalLightCurve_WASP-43 b_2021-12-27.png (f026a5bec7c1...), FinalLightCurve_WASP-43 b_2021-12-27.csv (c41c1fba7eb1...)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 01:43Z.